Undergraduate Topics in Mathematics 5

Abstract Algebra

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Contents

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1.1 Relations and functions

- **Relations**: Let S and T be sets. A **relation** from S to T is a subset ρ of $S \times T$, where

  $$S \times T = \{(s,t) : \ s \in S, t \in T\} .$$

  If $s \in S$ and $t \in T$, then we write $s\rho t$ if $(s,t) \in \rho$, otherwise we write $s\not\rho t$. In particular, a relation on S is a relation from S to S.

- **Properties of relations**: Let ρ be a relation on S,
  - **Reflexive**: ρ is reflexive if $a\rho a$ for all $a \in S$
  - **Symmetric**: ρ is symmetric if $a\rho b$ implies $b\rho a$
  - **Transitive**: ρ is transitive if $a\rho b$ and $b\rho c$ implies that $a\rho c$

  Note: These properties are defined only for relations from a set to itself.

- **Equivalence relation**: An **equivalence relation** on a set S is a relation that is reflexive, symmetric, and transitive. We use $\sim$ to denote an equivalence relation.

  As an example, on $\mathbb{R}$, let us say that $a \sim b$ if and only if $a + b$ is even. First, consider $a, b \in \mathbb{R}$. Observe that

  $$a \sim a$$

  since whether or not a is even or odd, the sum is always even. So, $\sim$ is reflexive. Next, we notice that if we assume $a \sim b$ (so $a + b$ is even), it immediately follows that $b + a$ is even. So, $b \sim a$ and $\sim$ is symmetric.

  Now, let $a, b, c \in \mathbb{R}$. Assume that

  $$a + b, \ b + c$$

  are both even. Hence,

  $$a + b + b + c = a + 2b + c = a + c + 2b$$

  is also even. Notice that $2b$ is even, so we consider two cases, either $a + c$ is odd and $\sim$ is not transitive or $a + c$ is even. If $a + c$ is odd, then we have $a + c = 2\ell + 1$, and

  $$a + c + 2b = 2\ell + 1 + 2b = 2(\ell + b) + 1,$$

  which is odd. But, we said that $a + c + 2b$ is even, a contradiction. Thus, $a \sim c$ and $\sim$ is transitive.

  Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation. ■

- **Equivalence class**: Let $\sim$ be an equivalence relation on a set S. If $a \in S$, then the **equivalence class** of a, denoted $[a]$, is the set $\{b \in S : \ a \sim b\}$

- **Partition**: Let S be a set, and let T be a set of nonempty subsets of S. We say that T is a **partition** of S if every $a \in S$ lies in exactly one set in T.

- **Theorem 1.1**: Let S be a set and $\sim$ be an equivalence relation on S. Then the equivalence classes with respect to $\sim$ form a partition of S. In particular, if $a \in S$, then $a \in [a]$, and $a \in [b]$ if and only if $[a] = [b]$.

Proof. Assume S is a set, and $\sim$ is an equivalence relation on S. To show that the equivalence relations with respect to $\sim$ form a partition of S, we must show that the equivalence relations are a set of nonempty pairwise disjoint subsets of S.

Let $a \in S$. By reflexivity of $\sim$, $a \sim a$. Thus, $a \in [a]$ and the equivalence classes are nonempty. Now, assume that

$$d \in [a] \cap [b].$$

Thus,

$$a \sim d, \quad b \sim d.$$

Now, if $c \in [a]$, then $a \sim c$. Notice that by transitivity,

$$d \sim a, \ a \sim c \implies c \sim d.$$

Similarly,

$$c \sim d, \ d \sim b \implies c \sim b.$$

Thus, $c \in [b]$. Since $c \in [a]$ implies that $c \in [b]$,

$$[a] \subseteq [b].$$

A similar argument, except starting with $c \in [b]$ yields that

$$[b] \subseteq [a].$$

Hence, if $d \in [a] \cap [b]$, it follows that $[a] = [b]$.

Therefore, if $a \in S$, then $a \in [a]$, and if $a \in [b]$, then $[b] = [a]$. Conversely, if $[a] = [b]$, then $a \in [a] = [b]$. ■

- **The partition**: So, now we know that for a equivalence relation $\sim$ on S,

  $$\sim \subseteq S \times S,$$

  if $a \in S$, then the equivalence class of a is the set

  $$[a] = \{b \in S : \ a \sim b\}.$$

  Further, $b \in [b]$, and so it must be that $[b] = [a]$. In other words, the representative element is not unique for each class.

- **Functions**: Let S and T be any sets. A **function**

  $$\alpha : \ S \to T$$

  is a rule assigning to each $s \in S$ an element $\alpha(s) \in T$.

- **Requirements of a function**: For a relation $f \subseteq A \times B$ to be called a **function** from A to B, it must satisfy

  1. **Existence / totality**: For all $a \in A$, there exists a $b \in B$ such that $(a, b) \in f$

     $$\forall a \in A, \ \exists b \in B \ : \ (a, b) \in f.$$

2. **Uniqueness / well-definedness**: Each input in A has exactly one output in B

$$\forall a \in A,\ \forall b_1, b_2 \in B,\ ((a, b_1) \in f \wedge (a, b_2) \in f) \implies b_1 = b_2.$$

- **Properties of functions**: Consider a function $\alpha : S \to T$,
  - **Injective**: α is injective if $\alpha(s_1) = \alpha(s_2)$ implies $s_1 = s_2$ for all $s_1, s_2 \in S$
  - **Surjective**: α is surjective if for all $t \in T$, there exits an element $s \in S$ such that $\alpha(s) = t$
  - **Bijective**: α is bijective (invertible) if it is both injective and surjective

  We call a bijective function a **bijection**, which is therefore also a **injection** and a **surjection**.

  Notice that the inverse of a function only satisfies existence / uniqueness if it is both injective (uniqueness) and surjective (existence).

- **Composition**: Let R, S, and T be sets, and let $\alpha : R \to S$ and $\beta : S \to T$ be functions. Then, the **composition** $\beta \circ \alpha$, or simply $\beta\alpha$, is the function from R to T given by

$$(\beta\alpha)(r) = \beta(\alpha(r))$$

  for all $r \in R$. Note that $\alpha\beta$ and $\beta\alpha$ are likely different functions.

- **Theorem 1.2: Properties of the composition**: Let $\alpha : R \to S$, $\beta : S \to T$, and $\gamma : T \to U$ be functions. Then,
  1. $(\gamma\beta)\alpha = \gamma(\beta\alpha)$
  2. If α and β are injective, then so is $\beta\alpha$
  3. If α and β are surjective, then so is $\beta\alpha$
  4. If α and β are bijective, then so is $\beta\alpha$

  Proof. First, notice that

$$(\gamma\beta)\alpha = \gamma(\beta(\alpha)),$$
$$\gamma(\beta\alpha) = \gamma(\beta(\alpha)).$$

  Hence, the first property holds. Now, consider $r_1, r_2 \in R$, and assume that

$$\beta(\alpha(r_1)) = \beta(\alpha(r_2)).$$

  Since β is injective, $\alpha(r_1) = \alpha(r_2)$, and since α is injective, $r_1 = r_2$. Thus, $\beta\alpha$ is injective.

  Next, consider $t \in T$. Since β is surjective, there exists a $s \in S$ such that $\beta(s) = t$. But, α is surjective, so there exists an $r \in R$ such that $\alpha(r) = s$. Thus,

$$(\beta\alpha)r = \beta(\alpha(r)) = \beta(s) = t.$$

  Since we showed that $\beta\alpha$ is both injective and surjective, it follows that $\beta\alpha$ is bijective. ■

- **Theorem 1.3: Existence of the inverse**: Let $\alpha : S \to T$ be a bijection. Then, there exists a bijection $\beta : T \to S$ such that $(\beta\alpha)(s) = s$ for all $s \in S$ and $(\alpha\beta)(t) = t$ for all $t \in T$.

Proof. Since α is bijective, for all $t \in T$, there exist a unique $s \in S$ such that $\alpha(s) = t$. Now, define $\beta : T \to S$ such that for that unique s that maps to t,

$$\beta(t) = s.$$

Then,

$$(\beta\alpha)(s) = \beta(\alpha(s)) = \beta(t) = s.$$

Now, for any $t \in T$, choose $s \in S$ such that $\alpha(s) = t$. By that same property of β,

$$\beta(t) = s,$$

so

$$(\alpha\beta)(t) = \alpha(\beta(t)) = \alpha(s) = t.$$

Now, we just need to show that β is a bijection. First, assume that $\beta(t_1) = \beta(t_2)$. Then,

$$\alpha(\beta(t_1)) = t_1,$$
$$\alpha(\beta(t_2)) = t_2.$$

But, $\beta(t_1) = \beta(t_2)$, so

$$\alpha(\beta(t_2)) = t_2 = \alpha(\beta(t_1)) = t_1.$$

Thus, β is injective. Next, $s \in S$, we have that

$$\beta(\alpha(s)) = s.$$

Thus, there exists a $t \in T$ such that $\beta(t) = s$ $(t = \alpha(s))$, so β is surjective. ■.

- **Permutation**: A permutation of a set S is a bijection from S to S.
- **Binary operation**: Let S be a set. Then, a **binary operation** on S is a function from $S \times S$ to S.

1.2 The Integers and Modular Arithmetic

1.2.1 Induction

- **Property 2.1: Well ordering axiom**: If S is a nonempty set of positive integers, then S has a smallest element.
- **Theorem 2.1: Mathematical induction**: Suppose that for each positive integer n, we have a proposition $P(n)$. Further suppose that

  1. $P(1)$ is true, and
  2. for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$

  Then, $P(n)$ is true for every positive integer n.

  Proof. We proceed by contradiction. Recall that to show a statement $P \implies Q$ is true, it is merely sufficient to show that $P \wedge \neg Q$ is impossible. Thus, if we assume $\neg Q$, and yield a contradiction, $P \wedge \neg Q$ cannot occur and thus the statement is true.

  So, we assume that $P(n)$ is not true for every positive integer n. Thus, there is a nonempty set $S \subseteq \mathbb{N}$ such that for every element $n \in S$, $P(n)$ is false. By the well-ordering axiom, there exists a smallest element in S, call this element k.

  Observe that the statement assumes that $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$. So, $k > 1$. Further,

  $$k - 1 \notin S,$$

  since if $k-1 \in S$, $k-1+1 = k \notin S$, but k is indeed a member of S. However, since $k-1 \notin S$, $P(k-1)$ is true, and thus $P(k-1+1) = P(k)$ is true, a contradiction.

  Therefore, if $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, $P(n+1)$ is also true, then $P(n)$ is true for every positive integer n. ■

  Let's consider an example. Suppose we claim that for every positive integer n,

  $$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

  First, we observe that

  $$1^2 = \frac{1(1+1)(2(1)+1)}{6} = 1.$$

  So, $P(1)$ is true. Next, assume that $P(n)$ is true, where

  $$P(n) = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

  Now, observe that

  $$\begin{aligned} P(n+1) &= 1^2 + 2^2 + \cdots + n^2 + (n+1)^2 = P(n) + (n+1)^2 \\ &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{n(2n+1)}{6} + (n+1)^2. \end{aligned}$$

1.2.2 Some Number Theory